

BOTOU, China

WMO: 546180

Lat: 38.083N

Lon: 116.550E

Elev: 43

StdP: 14.67

Time Zone: 8.00 (E08)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.0	16.4	-12.7	2.8	25.3	-8.5	3.5	25.7	14.4	31.5	12.6	32.2	2.8	290	0.411

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.5	95.8	73.9	93.3	74.2	91.0	73.9	81.8	89.2	80.2	87.5	78.7	85.7	6.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.8	155.0	86.5	78.2	146.6	85.1	76.6	139.0	83.5	45.8	89.1	44.0	87.9	42.4	85.8	86.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.6	12.3	10.7	DB	8.3	100.1	4.2	2.4	5.3	101.8	2.9	103.2	0.5	104.5	-2.5	106.3	(4)
(5)			WB	6.3	83.5	4.0	1.8	3.4	84.8	1.0	85.8	-1.2	86.8	-4.1	88.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	57.4	28.3	34.9	47.4	59.6	70.5	78.8	81.8	79.1	71.7	59.6	43.9	31.9	(6)	
	DBStd	19.75	5.94	6.99	8.16	7.24	6.10	5.50	4.26	4.22	5.48	6.77	7.72	5.88	(7)	
	HDD50	2035	673	423	147	9	0	0	0	0	0	8	213	562	(8)	
	HDD65	4606	1138	843	548	192	20	1	0	0	0	191	634	1027	(9)	
	CDD50	4737	0	0	66	296	636	864	986	903	651	306	29	0	(10)	
	CDD65	1832	0	0	2	30	190	415	521	438	213	23	0	0	(11)	
	CDH74	18552	0	0	18	267	1768	4586	5972	4277	1528	136	0	0	(12)	
	CDH80	7630	0	0	2	53	603	2186	2734	1616	420	16	0	0	(13)	
	WSAvg	4.9	4.1	4.9	6.2	6.6	5.9	5.4	4.6	4.1	4.2	4.5	4.3	3.9	(14)	
	PrecAvg	21.4	0.1	0.3	0.4	0.9	1.4	3.1	6.8	4.9	1.6	1.2	0.6	0.1	(15)	
(16) Precipitation	PrecMax	29.5	0.9	1.4	1.6	2.4	3.0	8.1	13.3	11.3	3.9	7.3	2.4	0.6	(16)	
	PrecMin	11.0	0.0	0.0	0.0	0.0	0.2	0.7	1.1	0.9	0.2	0.0	0.0	0.0	(17)	
	PrecStd	5.2	0.2	0.4	0.4	0.7	0.8	1.6	3.1	2.5	1.1	1.3	0.6	0.2	(18)	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.2	59.4	76.0	85.6	94.0	99.3	99.3	94.7	90.0	82.1	68.3	51.3	(19)	
		MCWB	36.6	44.4	56.1	63.6	67.4	70.1	74.5	81.4	73.0	64.8	53.7	39.4	(20)	
	2%	DB	43.9	54.5	70.8	80.6	89.5	96.2	95.9	91.7	87.2	77.8	63.3	46.9	(21)	
		MCWB	33.8	41.7	53.0	61.5	66.0	70.4	76.9	78.5	70.6	61.3	51.1	37.0	(22)	
	5%	DB	40.7	50.7	66.0	76.8	86.2	93.6	93.4	89.5	84.8	74.3	59.4	43.8	(23)	
		MCWB	32.0	39.6	50.6	59.2	64.9	69.9	77.1	77.2	69.5	59.6	49.0	35.4	(24)	
	10%	DB	37.7	46.7	61.3	73.2	83.2	90.9	90.9	87.6	82.2	70.8	55.6	40.9	(25)	
		MCWB	30.4	37.1	47.3	57.1	63.9	69.6	76.6	76.3	68.0	57.9	46.7	34.1	(26)	
	0.4%	WB	37.8	47.4	60.2	68.7	72.5	77.6	83.7	84.1	78.2	68.2	56.0	42.2	(27)	
		MCDB	46.3	55.1	71.4	80.8	85.4	89.8	91.5	91.2	86.4	75.5	64.2	47.8	(28)	
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	34.9	43.4	55.5	64.6	70.6	75.6	82.1	81.9	74.5	65.5	53.1	39.1	(29)	
		MCDB	42.1	51.4	67.4	76.4	84.1	87.6	90.0	88.5	82.5	72.4	60.0	44.0	(30)	
	5%	WB	33.0	40.6	51.9	61.9	68.8	74.1	80.6	80.3	72.5	63.0	50.6	36.8	(31)	
		MCDB	38.9	48.4	63.5	73.3	81.5	85.5	88.8	86.7	80.4	69.9	57.4	42.0	(32)	
	10%	WB	31.3	38.3	48.4	59.2	66.8	72.7	79.3	78.8	70.7	60.6	48.2	34.9	(33)	
		MCDB	36.4	46.0	59.9	70.3	78.7	83.9	87.0	84.7	78.6	67.7	54.2	39.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	15.5	16.4	18.7	19.6	19.4	18.7	14.5	13.7	16.3	17.4	16.0	14.7	(35)	
		MCDBR	19.1	21.6	23.7	23.7	23.8	22.6	18.1	15.5	19.1	20.9	20.1	18.6	(36)	
		MCWBR	12.1	12.4	11.9	11.0	8.7	6.8	5.9	6.0	6.9	9.0	11.0	11.2	(37)	
	5% WB	MCDBR	16.6	19.4	22.3	21.6	20.0	18.1	14.3	13.5	15.8	16.5	17.7	15.7	(38)	
		MCWBR	11.1	12.2	12.3	12.2	9.5	7.0	6.5	6.5	7.3	8.9	10.8	10.4	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.502	0.574	0.586	0.632	0.671	0.810	0.796	0.762	0.689	0.668	0.562	0.505	(40)	
	taud		1.778	1.665	1.653	1.604	1.574	1.394	1.460	1.502	1.576	1.571	1.715	1.808	(41)	
	Ebn at Noon		195	198	215	215	211	183	184	185	187	171	174	182	(42)	
	Edh at Noon		52	66	74	82	86	102	95	89	79	72	55	48	(43)	
	RadAvg		729	937	1386	1662	1860	1689	1510	1440	1288	1004	762	666	(44)	
(45) All-Sky Solar Radiation	RadStd		82	131	124	100	134	146	102	118	118	102	115	83	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(46)
(47)	Regional (11 neighbors)	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(47)

Nomenclature: See separate page